Hardware-light neural response estimation for naturalistic video: a reproducible protocol using an open multimodal brain-encoding foundation model

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Abstract

Traditional neuromarketing and, more broadly, stimulus–response neuroscience rely on electroencephalography (EEG) or functional magnetic resonance imaging (fMRI), which are costly, slow, and hard to scale beyond a handful of participants. Recent multimodal *brain-encoding foundation models* predict cortical responses to naturalistic audio–visual stimuli directly from the stimulus, offering a "hardware-light, software-heavy" alternative that requires no scanner and no participant at inference time. We present a *reproducible protocol* that operationalises one such model—the open-source TRIBE v2 trimodal brain encoder—for estimating vertex-wise cortical activity from short naturalistic video, using free cloud computation (Google Colab, one T4 GPU). We document the full pipeline: video ingestion, automatic speech transcription, per-modality feature extraction, and projection onto the `fsaverage5` cortical surface (20,484 vertices) at a one-second sampling interval. As a feasibility demonstration we first reproduce the model's documented reference behaviour on a standard trailer clip, then apply the protocol to three fast-moving-consumer-goods (FMCG) television advertisements. We report the pipeline's compute characteristics and qualitative output, and we are explicit about the demonstration's boundaries: the runs were bimodal (video and audio; the language pathway was inactive), the predicted activity arrays were not persisted, and no ground-truth neural data were available, so we make no quantitative accuracy claim. To close that gap we specify a pre-registered validation protocol—self-report affect ratings and public engagement metrics correlated against persisted per-region predictions—together with a power analysis. The contribution is a transparent, transferable method and an open, auditable recipe for scanner-free neural response estimation, applicable well beyond advertising to clinical naturalistic-stimulus paradigms.

Keywords: brain encoding, foundation models, neuro-simulation, naturalistic stimuli, digital phenotyping, reproducibility, explainable AI, open science

1 Introduction

Understanding how the human brain responds to rich, naturalistic stimuli is a central goal of both cognitive neuroscience and its applied offshoots. In consumer neuroscience—"neuromarketing"—the question is narrower but concrete: which moments of an advertisement engage a viewer, and how strongly [1, 14]. The dominant measurement tools, EEG and fMRI, are powerful but expensive, slow, and difficult to scale: a single fMRI session can cost hundreds to thousands of units of currency, recruitment and scheduling take weeks, and sample sizes rarely exceed a few dozen participants [15]. These constraints are not unique to marketing; they limit any programme that seeks to characterise brain responses to many stimuli, many times, at low marginal cost—precisely the regime that "precision neuroscience at scale" demands.

A complementary strategy has matured rapidly: rather than *measuring* the brain for every new stimulus, one can *predict* its response with an *encoding model* learned from prior recordings [9, 12]. Classical encoding models map hand-designed or network-derived stimulus features onto voxel- or vertex-wise activity. The current generation replaces bespoke features with large pretrained representations and couples them to the cortical surface through a shared transformer, yielding *brain-encoding foundation models* that generalise across stimuli and, to a degree, across people. TRIBE v2 (Trimodal Brain Encoder) is a recent open example: it fuses video, audio, and text representations—from V-JEPA 2 / DINOv2, Wav2Vec-BERT, and Llama 3.2 respectively—and predicts fMRI activity across the whole cortical surface for naturalistic movie stimuli [2, 4, 7, 11, 13]. Because inference needs only the stimulus and a GPU, such models turn a hardware problem into a software one.

This "hardware-light, software-heavy" shift is exactly the opening that the present Research Topic identifies: automated, secure, scalable tools for brain monitoring and analysis, built from open software pipelines rather than new instrumentation. Our aim here is deliberately methodological. We do *not* claim a new model, and we do *not* claim validated neuromarketing findings. Instead we contribute:

1. A **reproducible, low-cost protocol** for estimating vertex-wise cortical responses to short naturalistic videos using an open brain-encoding foundation model on free cloud hardware, specified precisely enough to re-run (exact package pins, model snapshot, and step-by-step procedure; Section 2).
2. A **transparent feasibility demonstration** on three FMCG advertisements, benchmarked against the model's documented reference behaviour, with an honest account of what the runs did and did not establish (Section 4).
3. A **pre-specified validation protocol**—self-report affect and public engagement metrics against persisted per-region predictions, with a power analysis—that any group can execute to convert the demonstration into a quantitative study (Section 3).

We frame advertising merely as a convenient, license-clear stimulus domain. The same protocol applies to any naturalistic-video paradigm, including the clinical assessment settings—using film as a standardised cognitive probe—that motivate this Research Topic (Section 5).

2 Materials and Methods

The protocol takes a short video file and returns a matrix of predicted cortical activity of shape $T \times V$, where T is the number of one-second time steps (TRs) and $V = 20{,}484$ is the number of vertices on the `fsaverage5` surface. Figure 1 summarises the flow.

2.1 Model

We use **TRIBE v2**, an open multimodal brain-encoding model distributed through the Hugging Face Hub as `facebook/tribev2` [11]. The model was loaded from checkpoint `best.ckpt` (709 MB; snapshot `f894e783020944dcd96e5568550afe2aa9743f9f`) via the project's `TribeModel.from_pretrained` utility, with no modification to weights or architecture. TRIBE v2 ingests three

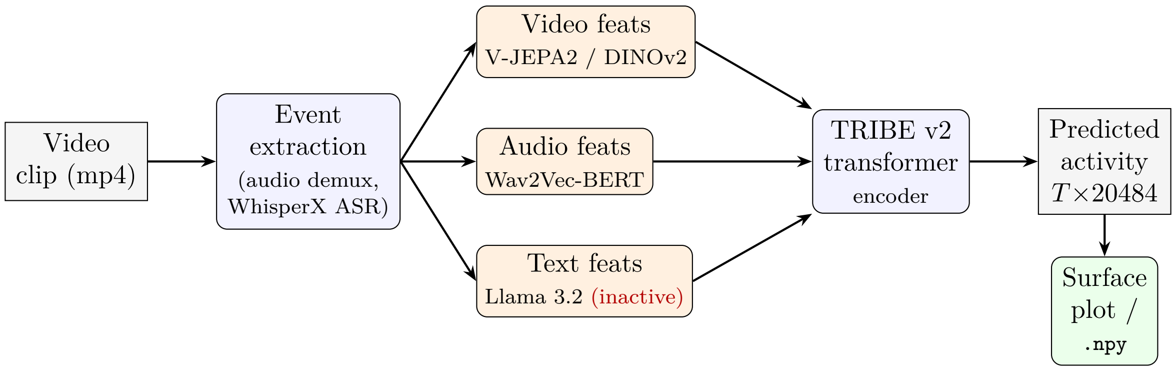


Figure 1: Protocol overview. A video clip is decomposed into time-stamped events; per-modality encoders produce feature streams that the TRIBE v2 transformer maps onto the `fsaverage5` cortical surface at 1 s resolution. In the demonstration runs the text/language pathway (dashed, red) was inactive because word-level events were filtered out before prediction (Section 2.5). The predicted-activity matrix should be persisted (`.npy`) to enable quantitative analysis; in the present runs only the rendered surface plots survive.

modalities through pretrained encoders— video via V-JEPA 2 and DINOv2, audio via Wav2Vec-BERT, and text via Llama 3.2 [2, 4, 7, 13]—and predicts blood-oxygen- level-dependent (BOLD) activity for each of $V = 20{,}484$ vertices of the `fsaverage5` template surface [6] at a repetition time of TR = 1 s. Following the model's convention, predictions are aligned to the stimulus with a 5 s offset to compensate for the hemodynamic lag. Speech is transcribed to word-level events with WhisperX [3, 16].

2.2 Compute environment and reproducibility

All runs were performed on Google Colab with a single NVIDIA T4 GPU. The software environment is pinned as follows (harvested from the executed notebook logs), which we report so the runs can be reproduced exactly:

- `tribev2==0.1.0` installed from source at commit `af58661791a351a448a489042a28f6c37e1c14b7`;
- `torch==2.6.0`, `torchvision==0.21.0`, `torchaudio==2.6.0` (force-reinstalled to match), with the CUDA 12.4 runtime stack;
- `x-transformers==1.27.20`, `triton==3.2.0`, `torchmetrics==1.9.0`, `submitit==1.5.4`;
- plotting stack `pyvista==0.48.4`, `vtk==9.6.2`.

The install pins `pandas==3.0.3`, which is newer than some preinstalled Colab packages expect (`cudf`, `dask-cudf`, `gcsfs` emit dependency warnings); these warnings did not affect the pipeline. The install command was

```
uv pip install \
    "tribev2[plotting] @ git+https://github.com/facebookresearch/tribev2.git"
pip install torchaudio==2.6.0 --force-reinstall --quiet
```

Two GPU-resident feature encoders dominate memory: the audio encoder (2.32 GB) and the video encoder (4.14 GB).

2.3 Stimulus set

The demonstration uses three publicly broadcast Indian FMCG television advertisements, chosen to span product categories and audio styles (Table 1): a Lay's potato-chips spot ("More smiles per mile"), a Colgate Visible White toothpaste spot (Hindi voice-over), and a Surf Excel detergent spot ("Primary Colours"). Each was downloaded at 360p and supplied to the pipeline as an `mp4`. As a methodological control we also reproduce the model's documented reference run on the Sintel open-movie trailer [5]. We use advertising because such clips are short, richly multimodal, and freely available; no human participants were recruited or recorded, so the demonstration carries no personal neural data (Section 5.4).

2.4 Procedure

For each clip the protocol executes five steps:

1. **Ingest.** Load the `mp4`; the toolkit demultiplexes the audio track to `wav`.
2. **Build events.** `model.get_events_dataframe(video_path)` produces a time-stamped event table containing `Video` and `Audio` spans plus WhisperX-derived `Word`/`Sentence`/ `Text` events.
3. **Select events.** The demonstration retained only `Video` and `Audio` events `df[df['type'].isin(['Video','Audio'])]`; see Section 2.5.
4. **Predict.** `preds, segments = model.predict(events=df)` returns activity of shape $T \times 20484$ and the corresponding time segments.
5. **Visualise.** `PlotBrain(mesh="fsaverage5")` renders the first 15 TRs on the lateral cortical surface (`cmap="fire"`, `norm_percentile=99`), with the stimulus frame shown above each panel.

2.5 A documented deviation, and the fix

Two properties of the demonstration runs must be stated plainly because they bound every downstream interpretation.

(i) The runs were bimodal, not trimodal. Step 3 above removed all word-level events, so no `Text` events reached the model and the Llama 3.2 language pathway was inactive. This is directly visible in the figures: the "Text" track is empty for all three advertisements (Figure 3) whereas it is populated for the trimodal reference run (Figure 2, words appearing at $t = 12$–$14\,\mathrm{s}$). For advertisements built around a spoken message—particularly the Hindi-language Colgate spot—suppressing the language pathway is a material limitation, not a detail. We report it as such and do not interpret any language-network activity.

(ii) The predicted arrays were not persisted. The runs rendered the surface plots but did not save `preds` to disk. Because the plots are percentile-thresholded, alpha-composited colour renders, the underlying $T \times 20484$ values cannot be recovered from them. Consequently *no* quantitative neural result (region-of-interest time courses, cross-ad comparisons, an "engagement index") can be computed from the existing outputs. The remedy is a single line after Step 4—

```
import numpy as np; np.save("preds.npy", preds)
```

—which we adopt as a mandatory step of the protocol going forward. Every quantitative analysis in Section 3 presupposes it.

3 Validation protocol (pre-specified)

Because the present runs yield no persisted predictions, we specify—rather than report—a validation study. Stating it in advance makes the eventual analysis falsifiable and pre-registrable.

Design. Re-run the protocol on a stimulus set of $K \geq 20$ advertisements with Step 4b (`np.save`) enabled and *all three modalities retained.* From each persisted $T \times 20484$ matrix derive two stimulus-level summaries per second: (a) mean predicted activity in an occipital/visual region of interest (ROI), and (b) mean activity in an auditory/temporal ROI, using an `fsaverage5` atlas parcellation. Aggregate to a per-advertisement engagement summary (e.g. time-averaged visual-ROI activity and its temporal variance).

External anchors. Collect two independent, low-cost criterion measures for the same advertisements:

- **Self-report.** A panel of N viewers rates each advertisement on validated valence and arousal scales (e.g. the Self-Assessment Manikin) plus a delayed brand-recall item.
- **Public engagement.** Platform metrics for the same creatives (e.g. view-through/retention rate, like-to-view ratio), or an established engagement dataset [10], as an ecological anchor.

Hypotheses and analysis. Test whether model-derived engagement summaries correlate with (i) self-reported arousal and (ii) public engagement, using Spearman correlations with bootstrap confidence intervals, and a permutation test that shuffles the advertisement–criterion pairing. Report effect sizes, not only p-values.

Power. For a target Spearman $\rho = 0.5$ at $\alpha = 0.05$ (two-sided) and power 0.80, roughly $K \approx 29$ advertisements are required; detecting a weaker $\rho = 0.4$ needs $K \approx 46$. The self-report panel size N is set so that each advertisement's mean rating has a standard error below a pre-specified threshold (e.g. $N \approx 30$ ratings per advertisement for typical rating variance). These numbers scope the effort before any claim is made.

4 Results: feasibility demonstration

We report what the runs establish—that the protocol executes end to end and produces spatially plausible output—and nothing beyond it.

4.1 Reference reproduction

Running the unmodified reference pipeline on the Sintel trailer reproduced the model's documented qualitative behaviour (Figure 2): early frames drive posterior (occipital, visual) cortex, and when the character begins to speak around $t = 12$–$14\,\mathrm{s}$—with word events present on the "Text" track—activity extends into more anterior and temporal territory consistent with the language network. Reproducing this reference behaviour before applying the pipeline to new stimuli is a cheap, informative sanity check that the environment, weights, and hemodynamic alignment are configured correctly.

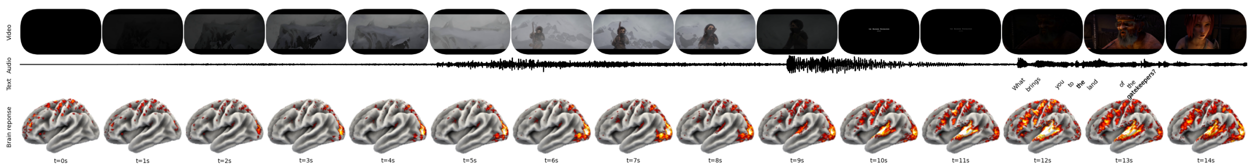


Figure 2: Reference reproduction (trimodal). Predicted cortical activity for the first 15 TRs of the Sintel trailer. Rows: stimulus frame, audio waveform, text (word events), and predicted activity on the lateral surface. Word events populate the "Text" track at $t = 12\text{–}14\,\text{s}$; posterior visual cortex responds to the visual stream throughout. This panel is the pipeline control, not a study result.

Table 1: Measured run characteristics (from executed notebook logs). T is the number of predicted one-second TRs; encoding time is the video feature-extraction stage on a single T4 GPU.

Stimulus	Duration (s)	Resolution	fps	T (TRs)	Video encoding
Sintel (reference)	52.21	854 × 480	24	53	29 min 26 s
Lay's	25.09	640 × 360	25	26	13 min 44 s
Colgate	20.13	640 × 360	25	21	11 min 14 s
Surf Excel	33.01	640 × 360	25	34	17 min 34 s

4.2 Application to advertisements

Applying the (bimodal) protocol to the three advertisements produced, in every case, activity concentrated in posterior occipital cortex that builds over the first several seconds of continuous visual stimulation (Figure 3). The Surf Excel spot elicited the most sustained and spatially extensive posterior activation across $t \approx 5\text{–}11\,\text{s}$; the Lay's and Colgate spots showed similar but somewhat less intense posterior responses. We describe these patterns *qualitatively only.* Because the prediction arrays were not persisted (Section 2.5) we compute no ROI statistics and draw no comparative or affective conclusions; the "Text" track is empty in all three panels, a visual confirmation that the language pathway did not contribute.

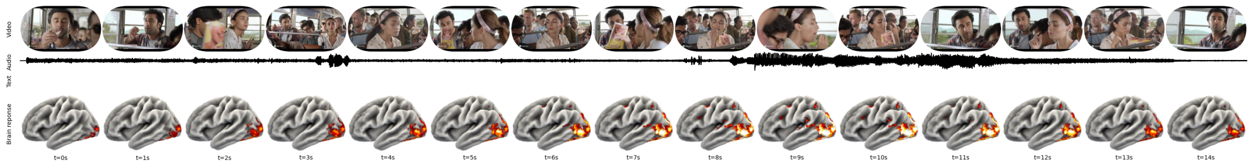
4.3 Pipeline characteristics

Table 1 reports the runs' measured properties. Two points warrant clarification. First, the number of predicted TRs equals the ceiling of the clip duration in seconds (e.g. 26 TRs for the 25.09 s Lay's clip): the model prepares up to 100 candidate one-second segments and retains those fully covered by the stimulus, so "segments kept" simply tracks clip length rather than indicating discarded data. Second, video feature extraction dominates wall-clock time (roughly 16 s per encoded window on a T4), so end-to-end time scales with clip duration—about 11–18 minutes for these 20–33 s advertisements, and 29 minutes for the 52 s reference clip.

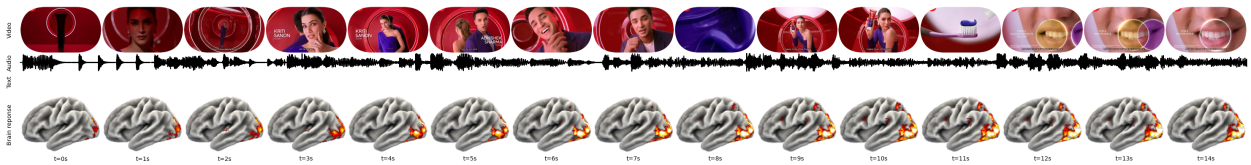
5 Discussion

5.1 What the demonstration shows

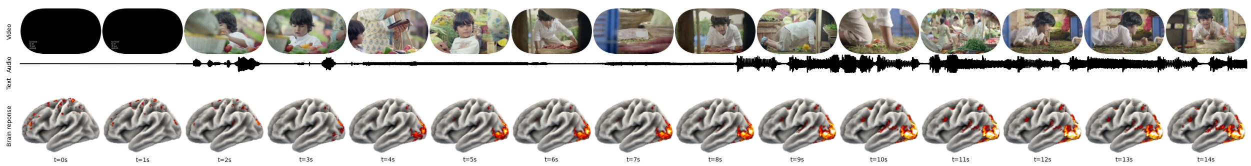
The protocol runs end to end on free cloud hardware and produces cortical response estimates whose gross spatial structure is plausible: visual cortex tracks the visual stream, and the trimodal



(a) Lay's — "More smiles per mile" (25.09 s)



(b) Colgate Visible White — Hindi (20.13 s)



(c) Surf Excel — "Primary Colours" (33.01 s)

Figure 3: Bimodal demonstration on three FMCG advertisements. Predicted cortical activity for the first 15 TRs. In all three, the "Text" track is empty (word events were filtered out before prediction) and activity is concentrated in posterior visual cortex. Panels are a feasibility demonstration, not validated measurements.

reference run additionally engages language-related cortex when speech is present. This establishes *feasibility* of scanner-free, participant-free neural response estimation for naturalistic video—the core promise of brain-encoding foundation models—using an entirely open toolchain that a small group can operate.

5.2 Transferability beyond advertising

Nothing in the protocol is specific to marketing. Any short naturalistic video can be substituted, which situates the method within the clinical and digital-phenotyping aims of this Research Topic: standardised film clips are already used as cognitive and affective probes [8], and a scanner-free estimate of the expected cortical response to such a probe could support hypothesis generation, stimulus selection, or triage before committing scarce imaging resources. We stress that clinical use would require the validation in Section 3 and, ultimately, comparison against measured neural data in the target population—neither of which we claim here.

5.3 Limitations

We are deliberately explicit:

- **No ground truth.** No EEG/fMRI or behavioural criterion was available, so predictive validity is untested.
- **Bimodal runs.** The language pathway was inactive (Section 2.5); results reflect video and

audio only.

- **No persisted arrays.** Only rendered plots survive, precluding quantitative analysis of the existing runs.
- **Small stimulus set and single "subject."** Three advertisements and the model's default subject identifier do not support generalisation claims.
- **Domain shift.** TRIBE v2 was trained on predominantly English-language, Western naturalistic media; applying it to Hindi-language Indian advertisements is an out-of-distribution use whose accuracy is unknown and plausibly reduced.

5.4 Explainability, bias, and privacy

The Research Topic foregrounds explainable AI, bias, and data security, and the method engages all three. **Interpretability:** encoding-model outputs are spatial and temporal, hence inspectable against known functional anatomy (the visual-cortex response is a face-valid check), but the mapping from stimulus features to vertices remains opaque and should be probed with perturbation and ablation analyses. **Bias:** the domain-shift limitation above is squarely an equity issue—a model trained on Western media may estimate less accurately for other languages and cultural contexts, and validation must be stratified accordingly. **Privacy:** a genuine advantage of the hardware-light approach is that inference collects *no* neural or personal data from any individual; the only human data enter through the optional, consented self-report panel of the validation protocol, which should follow standard ethical review.

5.5 Conclusion

We have presented a transparent, reproducible protocol for estimating cortical responses to naturalistic video with an open brain-encoding foundation model, demonstrated its feasibility on three advertisements against a reference control, and specified the validation needed to make quantitative claims. The value is methodological and infrastructural: an auditable, low-cost recipe that lowers the barrier to naturalistic stimulus–response neuroscience, and a candid map of the gap between a working pipeline and a validated measurement.

Reproducibility and data availability

The notebooks, the figure-extraction script (`paper/extract_figures.py`), and the recovered figures are in the project repository: `https://github.com/souricebunny/NeuroMarketing`. The model is `facebook/tribev2` on the Hugging Face Hub. No human-participant data were collected for this work.

Conflict of interest

The author declares that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Author contributions

R.G. is the sole author and conceived the study, designed and executed the computational pipeline, performed the analyses, recovered and prepared the figures, and wrote the manuscript. The author read and approved the submitted version. The work was carried out under the academic guidance of Dr Misha Kakkar.

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References

[1] Dan Ariely and Gregory S. Berns. Neuromarketing: the hope and hype of neuroimaging in business. *Nature Reviews Neuroscience*, 11(4):284–292, 2010. doi: 10.1038/nrn2795.

[2] Alexei Baevski, Henry Zhou, Abdelrahman Mohamed, and Michael Auli. wav2vec 2.0: A framework for self-supervised learning of speech representations. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. VERIFY. TRIBE v2 uses a Wav2Vec-BERT variant; cite the specific model if required.

[3] Max Bain, Jaesung Huh, Tengda Han, and Andrew Zisserman. WhisperX: Time-accurate speech transcription of long-form audio. In *Interspeech*, 2023. VERIFY.

[4] Adrien Bardes et al. V-jepa 2: Self-supervised video models for understanding and prediction. *arXiv preprint*, 2025. VERIFY arXiv ID and author list.

[5] Blender Foundation. Sintel (durian open movie project) trailer. `https://durian.blender.org/`, 2010. Reference stimulus; VERIFY licensing statement for reuse.

[6] Bruce Fischl. FreeSurfer. *NeuroImage*, 62(2):774–781, 2012. doi: 10.1016/j.neuroimage.2012.01.021.

[7] Aaron Grattafiori et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024. VERIFY author list and ID.

[8] Uri Hasson, Yuval Nir, Ifat Levy, Galit Fuhrmann, and Rafael Malach. Intersubject synchronization of cortical activity during natural vision. *Science*, 303(5664):1634–1640, 2004. doi: 10.1126/science.1089506.

[9] Alexander G. Huth, Wendy A. de Heer, Thomas L. Griffiths, Frederic E. Theunissen, and Jack L. Gallant. Natural speech reveals the semantic maps that tile human cerebral cortex. *Nature*, 532(7600):453–458, 2016. doi: 10.1038/nature17637.

[10] Karnika Kapoor. Video ads engagement dataset. Kaggle. https://www.kaggle.com/datasets/karnikakapoor/video-ads-engagement-dataset, 2021. VERIFY exact title/author/year on Kaggle.

[11] Meta AI (FAIR). TRIBE: Trimodal brain encoder for whole-brain fmri response prediction. Model and code: https://huggingface.co/facebook/tribev2; https://github.com/facebookresearch/tribev2; project page https://aidemos.atmeta.com/tribev2, 2025. VERIFY authorship/citation of the accompanying TRIBE paper (Algonauts 2025).

[12] Thomas Naselaris, Kendrick N. Kay, Shinji Nishimoto, and Jack L. Gallant. Encoding and decoding in fMRI. *NeuroImage*, 56(2):400–410, 2011. doi: 10.1016/j.neuroimage.2010.07.073.

[13] Maxime Oquab, Timothée Darcet, Théo Moutakanni, et al. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research (TMLR)*, 2024. VERIFY; arXiv:2304.07193.

[14] Hilke Plassmann, Vinod Venkatraman, Scott Huettel, and Carolyn Yoon. Consumer neuroscience: applications, challenges, and possible solutions. *Journal of Marketing Research*, 52 (4):427–435, 2015. doi: 10.1509/jmr.14.0048.

[15] Russell A. Poldrack et al. Scanning the horizon: towards transparent and reproducible neuroimaging research. *Nature Reviews Neuroscience*, 18(2):115–126, 2017. doi: 10.1038/nrn.2016.167. VERIFY full author list.

[16] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. In *International Conference on Machine Learning (ICML)*, 2023. VERIFY.